MA2108

Test 1

Wed. 16/09/2026

Time: 50 minutes Total marks: 25

Question 1. Let V be a vector space over a field F and W be a subset of V. What do you understand by:

a. 'W is a subspace of V over F'?

b. 'a set of vectors $\{u_1, u_2, ..., u_s\}$ is a basis of V'?

[2 + 2 = 4 marks]

Question 2. Let

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 5 \\ -1 \\ 4 \end{pmatrix}.$$

a. Is $\{v_1, v_2, v_3\}$ linearly independent?

b. Does $\{v_1, v_2, v_3\}$ span $\mathbb{R}^3$?

c. Find a basis for $U = \text{Span}\{v_1, v_2, v_3\}$.

d. What is $\dim(U)$?

[2 + 2 + 3 + 1 = 8 marks]

Question 3. For a vector space U with a basis $\{u_1, u_2, ..., u_s\}$, prove that for each $x \in U$, there exists a unique set of scalars $\alpha_1, \alpha_2, ..., \alpha_s$ such that

$$x = \alpha_1 u_1 + \alpha_2 u_2 + ... + \alpha_s u_s.$$

[6 marks]

Question 4. Which of the following are subspaces of $\mathbb{R}^4$? Justify your answer.

a. $U = \{(a, b, c, d)^T \in \mathbb{R}^4 : a + 2b = 0, c = d\}$.

b. $V = \{(a, b, c, d)^T \in \mathbb{R}^4 : cd = 0\}$.

[4 + 3 = 7 marks]

[4 + 8 + 6 + 7 = 25 marks]

Assoc. Prof. Dr N. Sookia.